MIL-A-23670A(AS)
Amendment - 3
12 October 1972

Superseding
Amendment - 2
22 February 1971

MILITARY SPECIFICATION

ATTITUDE HEADING REFERENCE SYSTEM
AN/ASN-50(*)

This amendment forms a part of Military Specification MIL-A-23670A(WP), dated 1 November 1965, and has been approved by the Naval Air Systems Command, Department of the Navy.

By this amendment basic MIL-A-23670A(WP) is changed to MIL-A-23670A(AS) and throughout the specification, Bureau of Naval Weapons shall be changed to Naval Air Systems Command.

Page 2

3.3.2.3, line 1: Delete "325" and substitute "400".

3.3.2.4: Delete and substitute

"3.3.2.4 Time Totalizing Meter - The equipment shall contain time totalizing meters in accordance with Specification MIL-M-7793. Meters shall be included in the following units:

Unit	Type of Meter
Amplifier-Power Supply	MS17322
Displacement Gyroscope	MS17322
Compass Adapter	MS17322"

Page 3

3.3.6, line 3 after MIL-I-6181: Add

"except that Short Duration Interference limit under paragraph 3.2.2 shall be 45 db for one second transient."

Page 10

3.5.1.1.5, lines 8 and 9: Delete and substitute

"When loaded with 8CT's, the line-to-line output voltage shall be 10.6 ±0.6V and the phase shift shall not exceed 10°."

FSC 6605

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Page 14

3.5.1.2.4, lines 9 and 10: Delete and substitute

"When loaded with 8CT's, the line-to-line output voltage shall be 10.6 ±0.6V and the phase shift shall not exceed 10°."

Page 17

FIGURE 4, dimension $\frac{11.010}{10.950}$: Delete and substitute

"$\frac{11.010}{10.850}$"

Page 20

3.5.2.8: Delete and substitute

"3.5.2.8 Performance at Environmental Extremes - When tested at environmental extremes, the amplifier-power supply shall either meet the tolerance specified for standard conditions or shall repeat within 20% the results of its individual tests made at standard conditions prior to the start of environmental tests, except that the servo amplifier maximum outputs shall be not less than 80% of specified outputs at standard conditions."

Page 29

Add the following new paragraphs:

* 3.5.6 Lubrication -

* 3.5.6.1 Rotor Bearing Lubrication - The rotor bearings shall be lubricated with grease conforming to MIL-G-81322 or MINIPURE grease with particle size not exceeding 50 microns.

* 3.5.6.2 Bearing Lubrication - The lubricating oil used in gimbal bearings shall be AS-2844 in accordance with the requirements of MIL-L-6085.

* 3.5.6.3 Barrier Film Bearing Coating - All bearings shall be processed in accordance with MIL-STD-1334. The coating material shall be in accordance with MIL-B-81744. All bearings, including chamfered races or corners, and including interfacing parts where applicable, shall be barrier coated.

* 3.5.6.4 Ball Bearings - Ball bearings shall be ABEC-5 grade or better. Teflon separators shall be used on gimbal bearings where the bearing design permits. Spin rotor bearings shall have porous separators impregnated with lubricant AS-2844 and shall be ABEC-7 or better.

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*3.5.6.5 Ball Bearing Pre-Loading - Means shall be provided for pre-loading and alignment without causing eccentric loading unless otherwise authorized.

Page 33

4.4.3, 4.4.3.1: Delete and substitute

"4.4.3 Reliability Assurance Tests - Reliability Assurance Tests shall be conducted using MIL-STD-781. Tests as required by both the Qualification Phase and the All Equipment test shall be conducted. Classification of failure shall be in accordance with MIL-STD-781 and AR-34."

"4.4.3.1 Qualification Phase - Prior to the acceptance of equipments under the contract or order, a minimum of three (3) equipments shall be tested as outlined in MIL-STD-781, under the section "Qualification Phase of Production Reliability Tests." The maximum number of equipments to be used shall be those listed in Table 5 of MIL-STD-781. The Accept-Reject Criteria for Test Plan III shall be used."

Add the following new paragraph:

*4.4.3.1.1 Reliability Assurance Production Acceptance Phase - Reliability Assurance Production Acceptance Testing shall be conducted through a production burn-in. Each equipment produced but not submitted to the Qualification Test shall be operated for 150 operating hours burn-in, and the test data (operating hours and failures) shall be totaled and compared to the Accept/Reject Criteria of Test Plan II of MIL-STD-781. (Extend the lines as necessary to accomodate the data.) These totals shall accumulate so that at any one time the experience from the beginning of the contract is included. At the conclusion of each month the test results shall be sent to the procuring activity and to the Naval Air Systems Command, Attention: Avionics Division. Delivery of equipment shall not be held up to await a decision under the test plan. In event an Accept Decision is reached, the test duration of each equipment shall be reduced to 100 operating hours. At any time that the current total of test hours and test failures plotted on Test Plan IV curves (extended) show a Reject Situation, the Procuring Activity shall be notified. The procuring Activity reserves the right to stop the acceptance of equipment at any time that a Reject Situation exists pending a review of the contractor's efforts to improve the equipment, the equipment parts, the equipment workmanship, etc., so that the entire compilation will show other than a rejection decision. For the Production Acceptance Phase, Test Level E shall be used, except that the temperature extremes shall be +55C and -44C, and vibration shall be 0.5G. Details of the burn-in tests shall be included in the test procedures.

Page 34

4.4.3.3: Delete in its entirety.

4.4.3.6: Delete "MIL-R-23094" and substitute "MIL-STD-781."

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Add

4.4.3.7 Test Details - The test details such as the length of the test cycle, the length of the heat portion of the cycle, the performance characteristics to be measured, special failure criteria, preventive maintenance to be allowed during the test, etc., shall be part of the test procedures of 4.6 to be submitted to and approved by the procuring activity prior to the beginning of the acceptance test phase of the reliability assurance tests.

Page 36

4.6, line 1, after "preproduction tests": Add

:, initial production tests".

Page 38

6.9 under Appl. Par.

"Delete	and	Substitute
3.4.1		3.5.1
3.4.2		3.5.2
3.4.3		3.5.3
3.4.4		3.5.4
3.4.5		3.5.5

Page 39

6.11: Delete

"MIL-R-23094 Reliability Assurance for Production Acceptance of Avionic Equipment, General Specification for."

and substitute

"MIL-STD-781 Reliability Tests, Exponential Distribution."

Add the following under military specifications:

MIL-L-6085	Lubricating Oil, Instrument, Aircraft, Low Volatility
MIL-G-81322	Grease, Aircraft, General Purpose Wide Temperature Range
MIL-B-81744	Barrier Coating Solution and Solvent

Add the following under military standards

MIL-STD-1334	Process for Barrier Coating of Anti-Friction Bearings

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Page 40

6.11 Delete MS17323 Meter, Time, Totalizing, Miniature 400 Cycle, Dial

At the end of para. 6.11, add the following:

Naval Air Systems Command

AR-34 Failure Classification for Reliability Testing, General Specification for

AS-2844 Lubricant, Instrument Ball Bearing, High Flash Point

Page 41

FIGURE 6, dimension $\frac{.177}{.176}$: Delete and substitute

$\frac{".177"}{.172}$

Specification Analysis Sheet

Delete "Bureau of Naval Weapons" and substitute "Naval Air Systems Command" (2 places)

Delete "(RREN-5)" and substitute "(AIR-53374B):

Project No. 6605-N233

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